

GATE Question solution using PlatformIO

Question

14) In the logic circuit shown in the figure, Y is given by

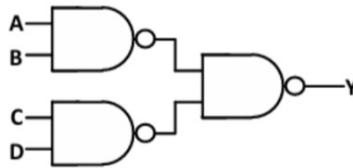


Fig. 4.

a) $Y = \bar{A}BCD$

b) $Y = (\bar{A} + B)(C + D)$

c) $Y = \bar{A} + B + C + D$

d) $Y = AB + CD$

Solution

The logic function given by the circuit is:

$$Y = AB + CD$$

This means the output Y is high when either both A and B are high or both C and D are high. The solution involves implementing this equation using Arduino and hardware (7447 IC). The function was verified by constructing the truth table and tested in the PlatformIO environment.

Truth Table

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Steps to Execute in PlatformIO

1. Install PlatformIO in your chosen development environment.
2. Initialize a new project for Arduino UNO:

```
pio project init --board uno
```

3. Place the Arduino code implementing the equation $Y = AB + CD$ in `src/main.cpp`.
4. Build the project:

```
pio run
```

5. On Android Termux, transfer the compiled firmware to ArduinoDroid:

```
cp .pio/build/uno/firmware.hex /sdcard/ArduinoDroid/sketch/sketch7447/sketch.hex
```

6. Upload the code via ArduinoDroid or use PlatformIO upload on PC.
7. Verify output on the 7-segment display connected to the 7447 driver IC.

Conclusion

This report demonstrated the implementation of the logic function $Y = AB + CD$ using Arduino and a 7447 BCD to 7-segment driver IC. The function was validated through a truth table and verified in hardware using PlatformIO workflow. This process confirms practical embedded system applications of combinational logic design.